



****DRAFT****

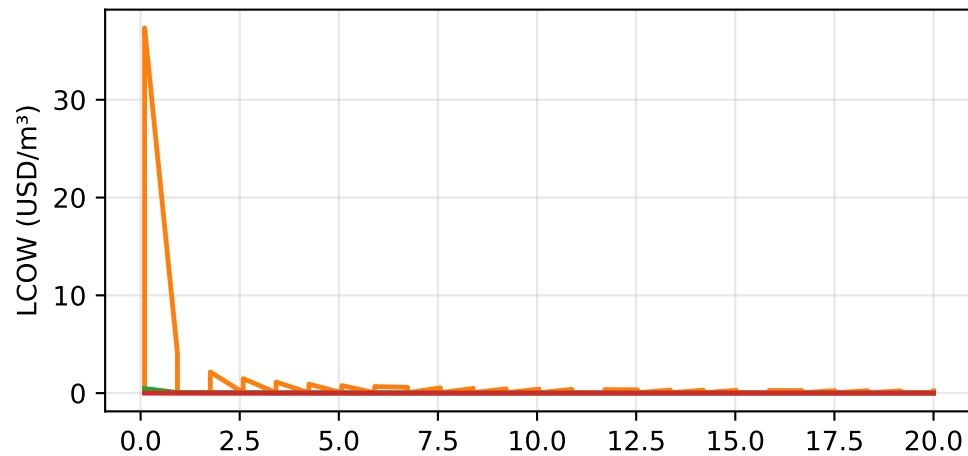
WaterTAP Zero-Order Model LCOW & CAPEX Curves

All costs in 2023 USD

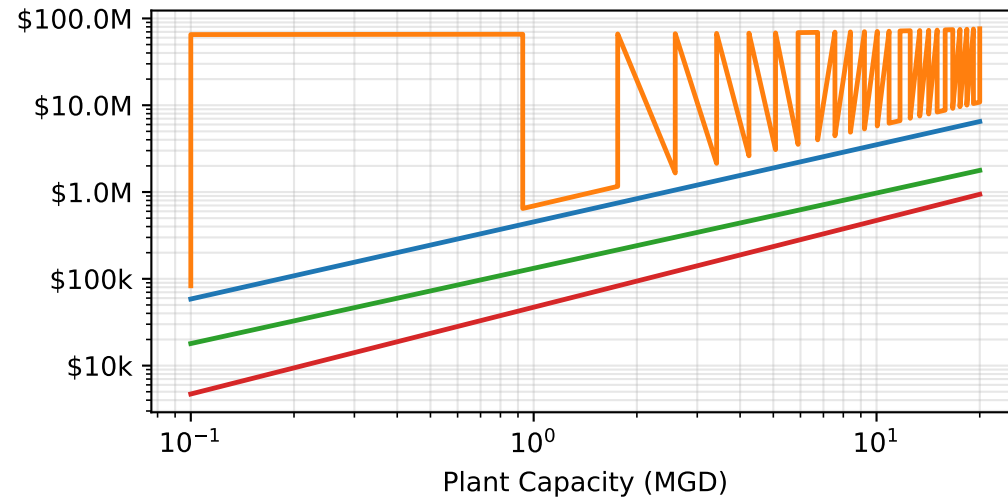
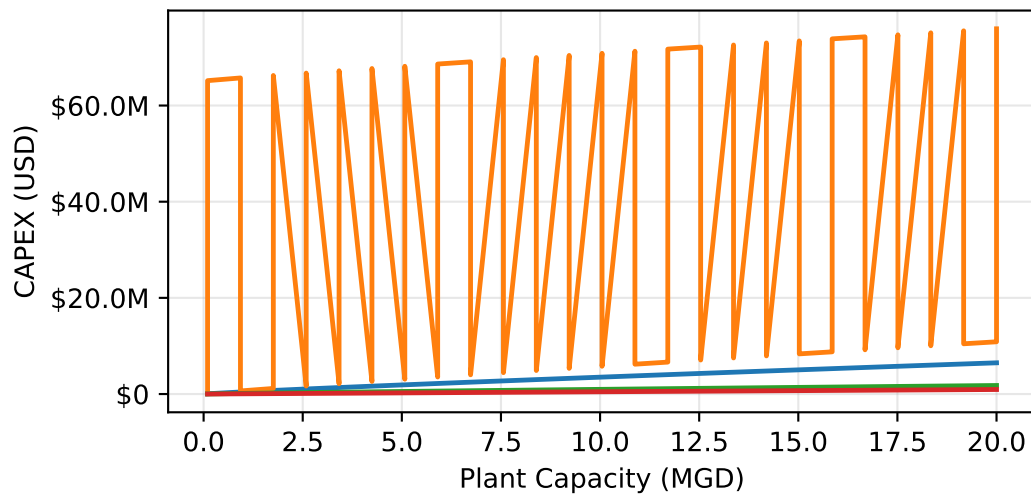
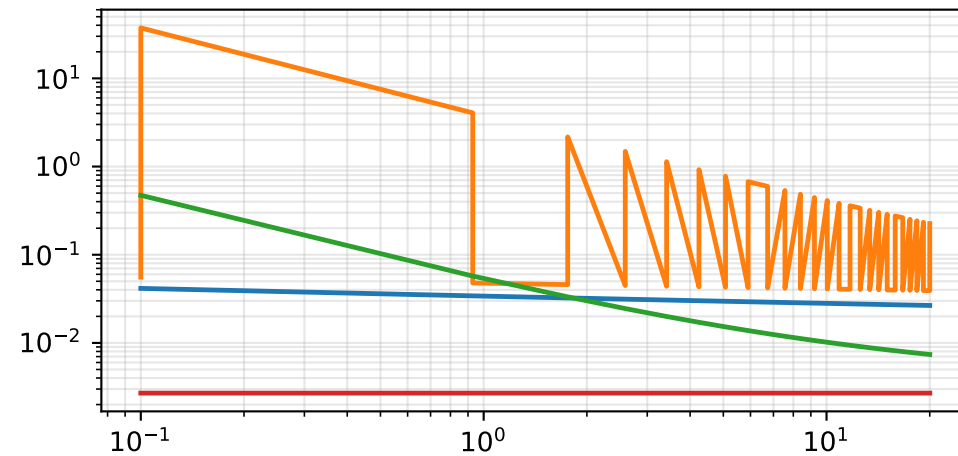
Generated from model runs

Intake & Conveyance

Linear Scale



Log-Log Scale

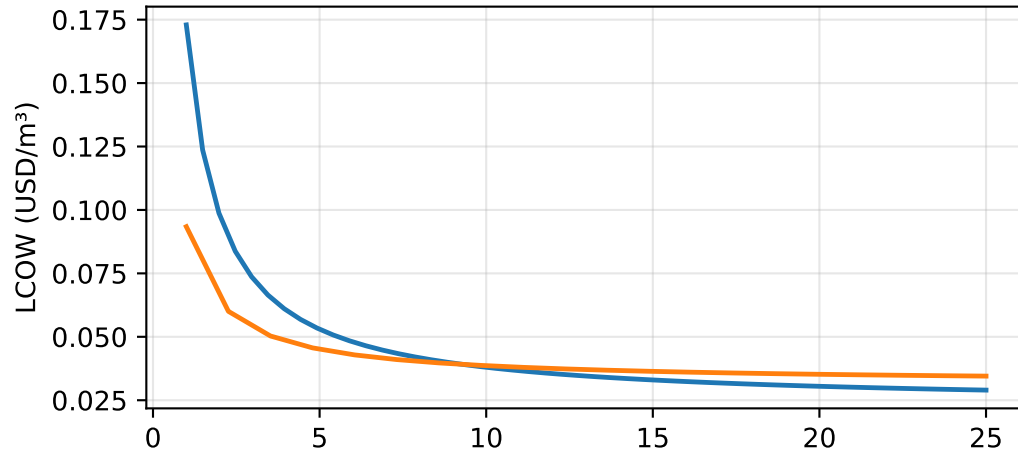


— Seawater Intake — Well Field — Water Pumping Station — Screen

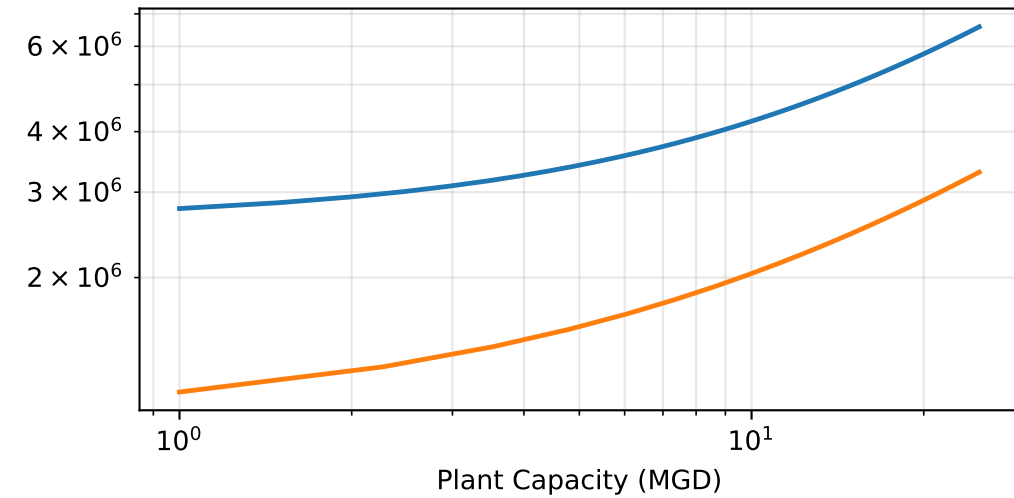
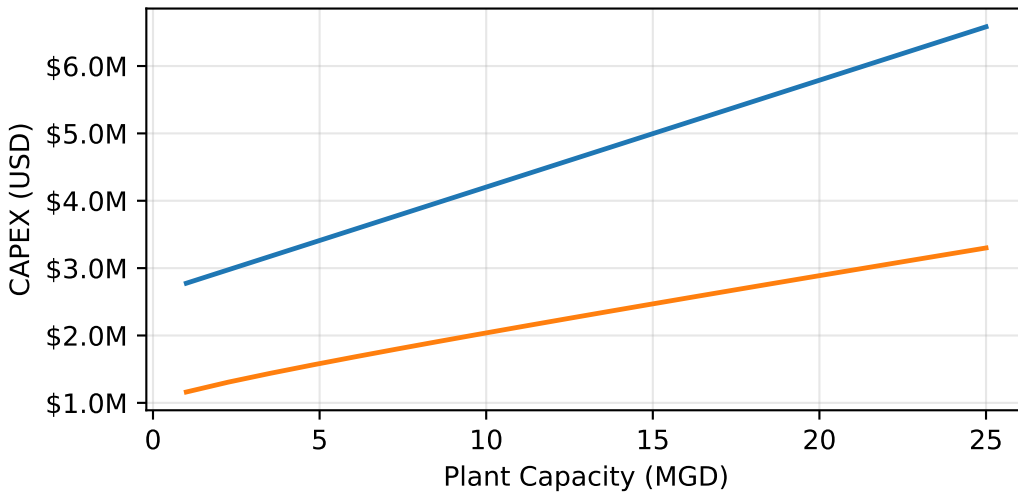
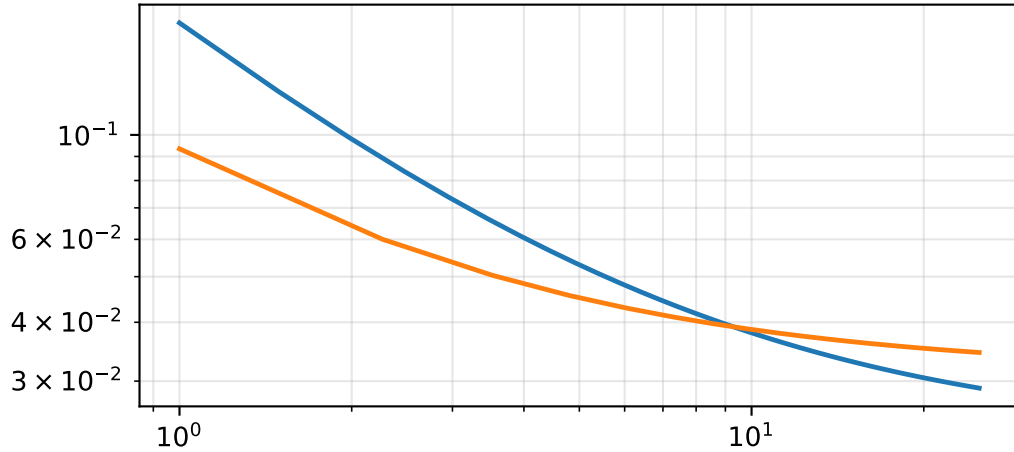


Pretreatment

Linear Scale



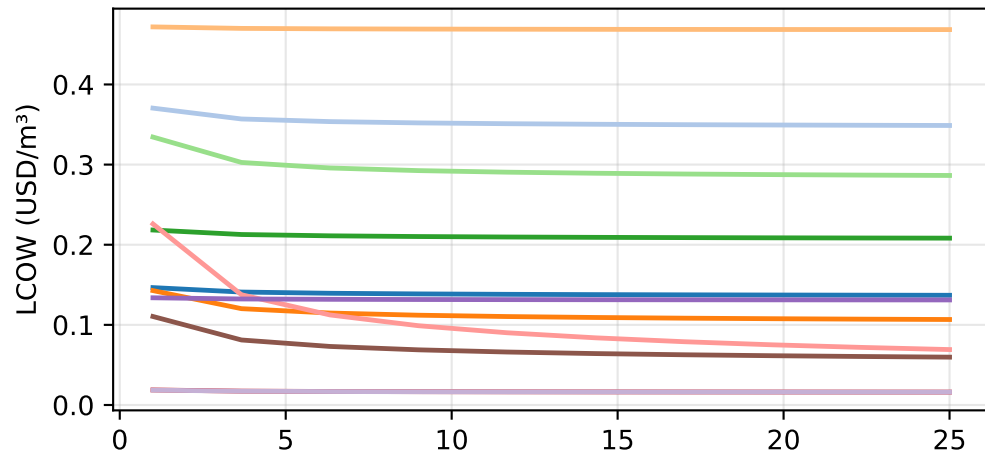
Log-Log Scale



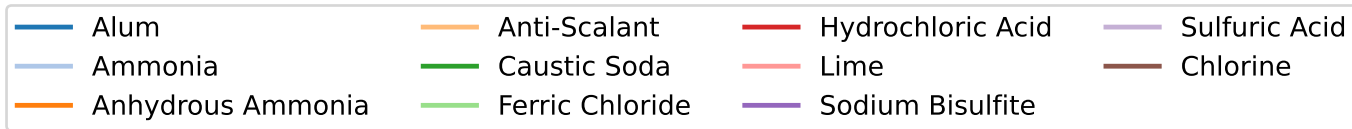
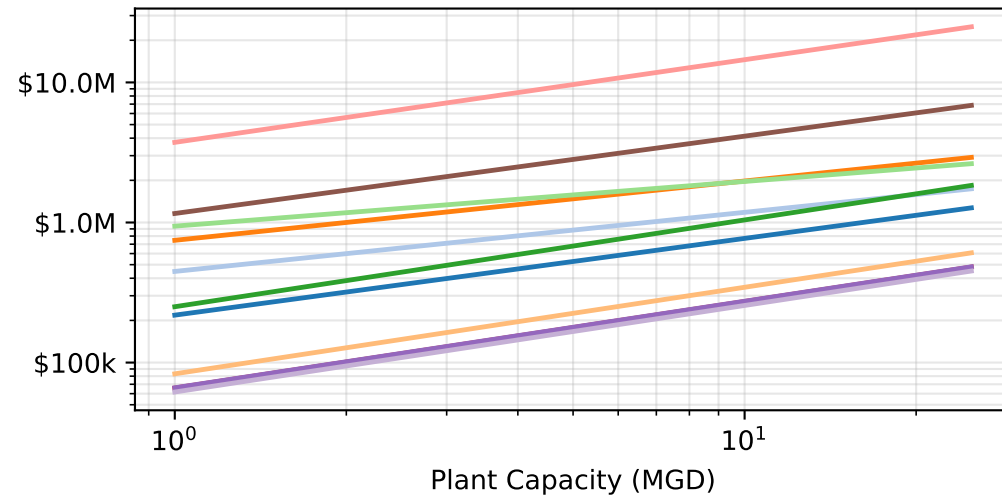
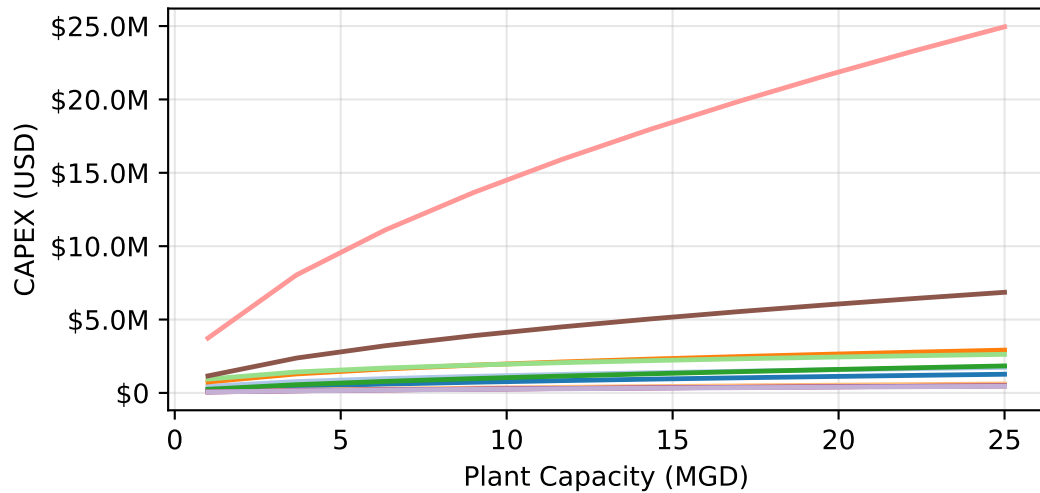
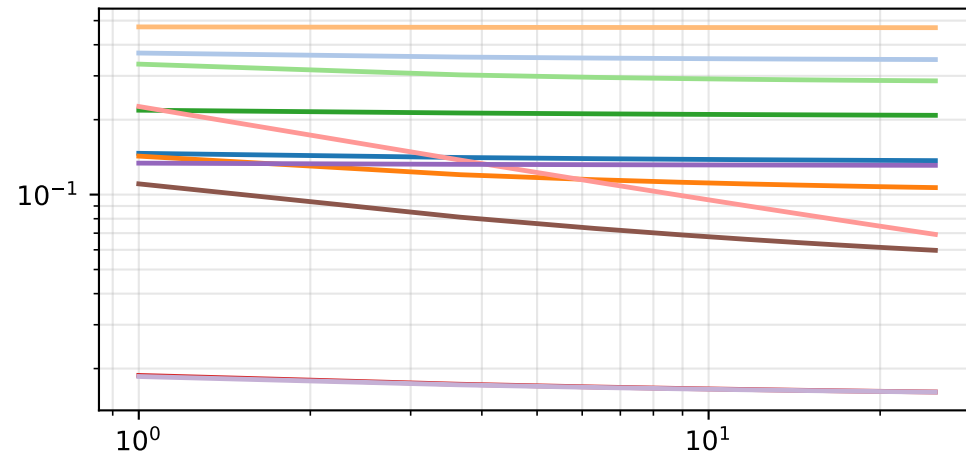
— Coagulation & Flocculation — Electrocoagulation

Chemical Addition

Linear Scale



Log-Log Scale

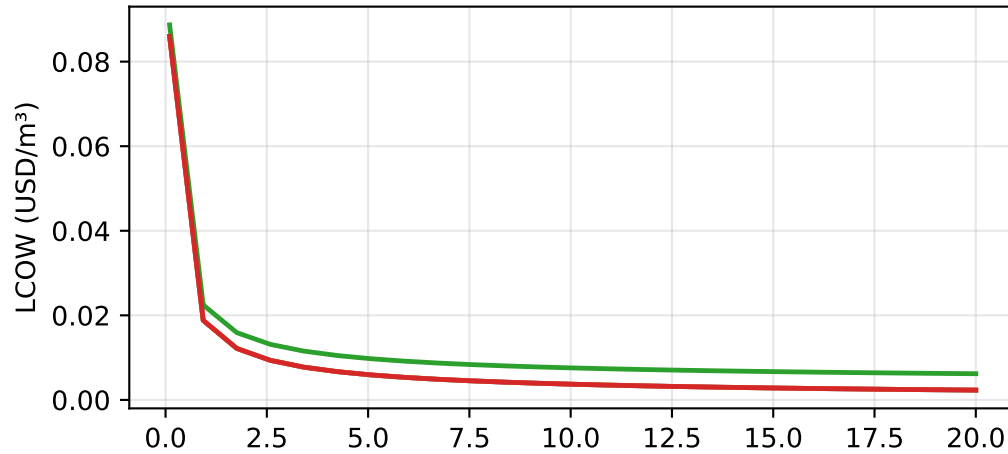


Details:
Chemical addition costs are run with a fixed dose of 100 mg/L for each chemical

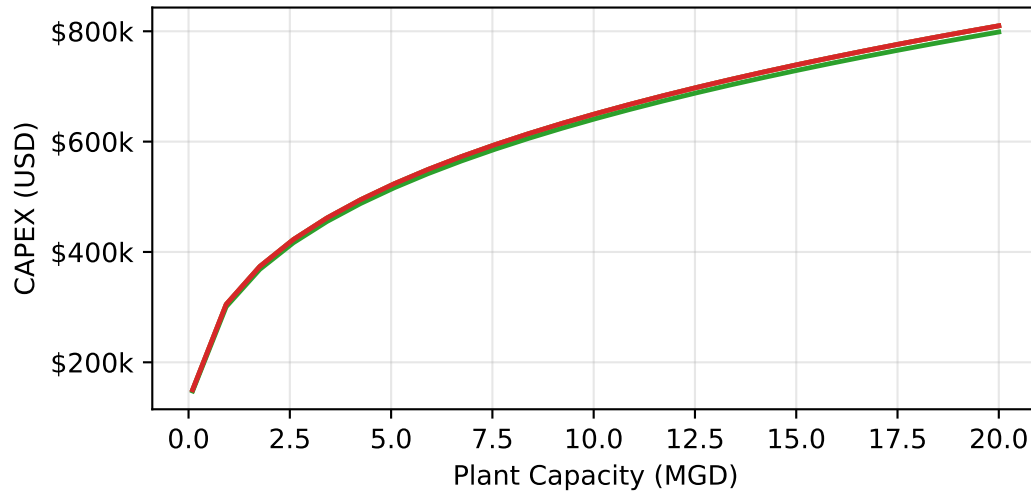
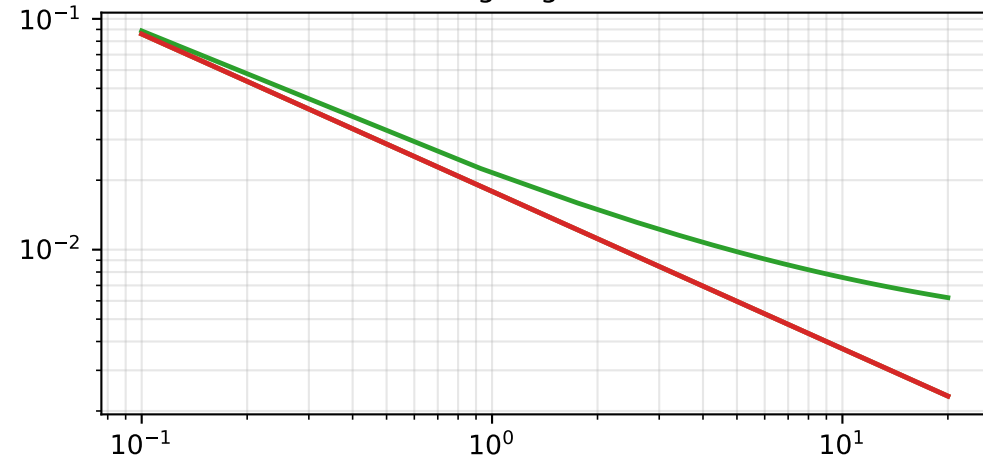


Clarification & Settling

Linear Scale



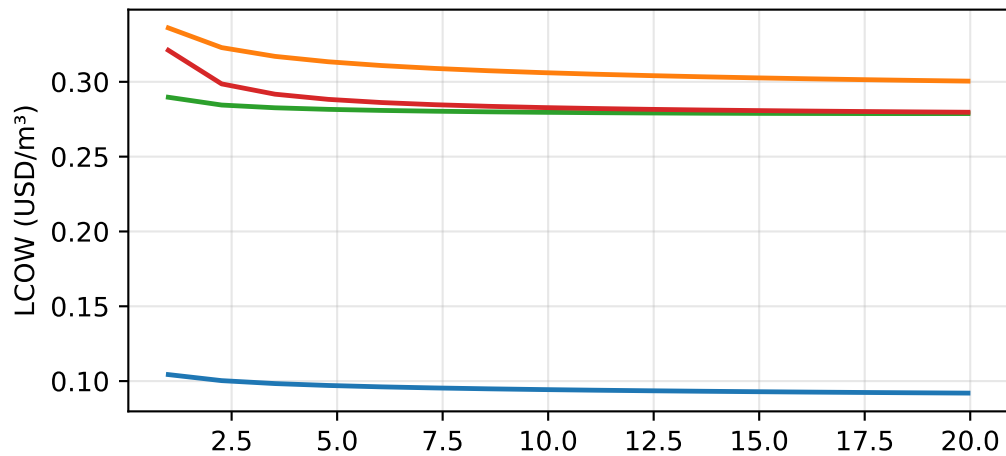
Log-Log Scale



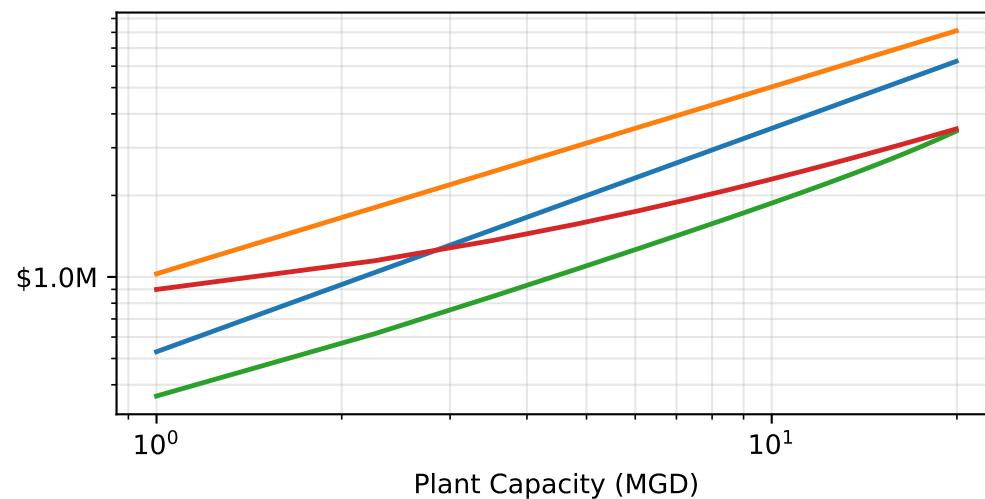
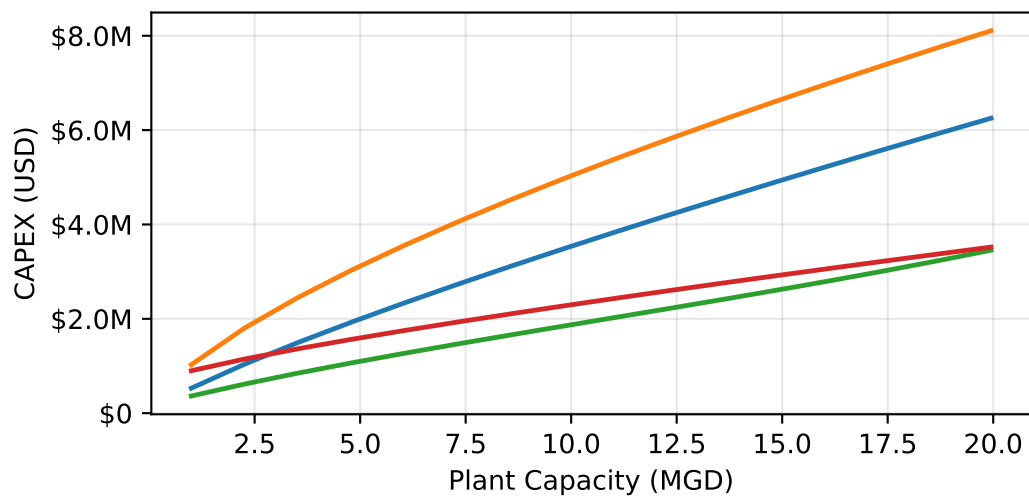
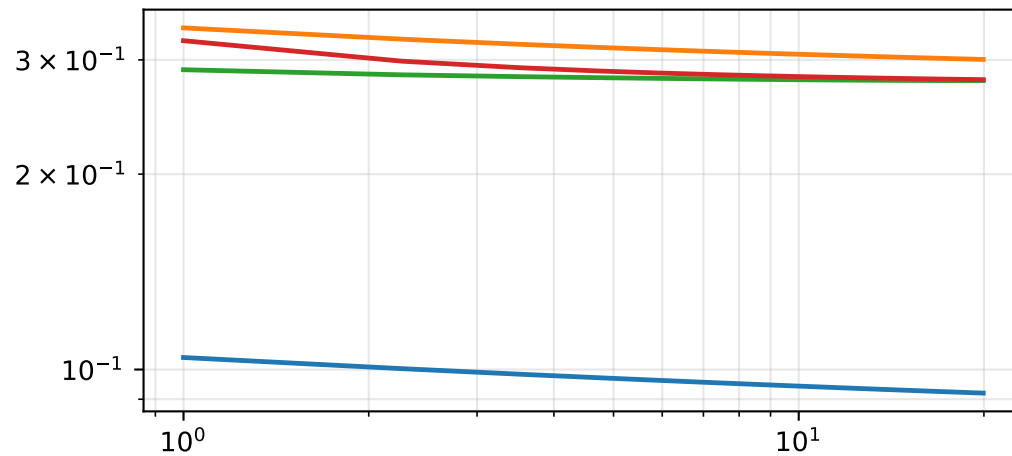
Primary Separator Settling Pond Dissolved Air Flotation Clarifier

GAC & Ion Exchange

Linear Scale



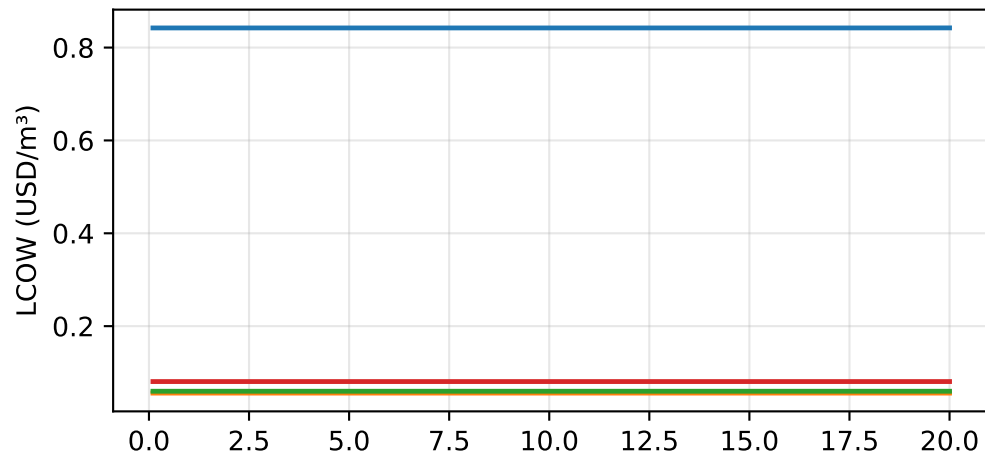
Log-Log Scale



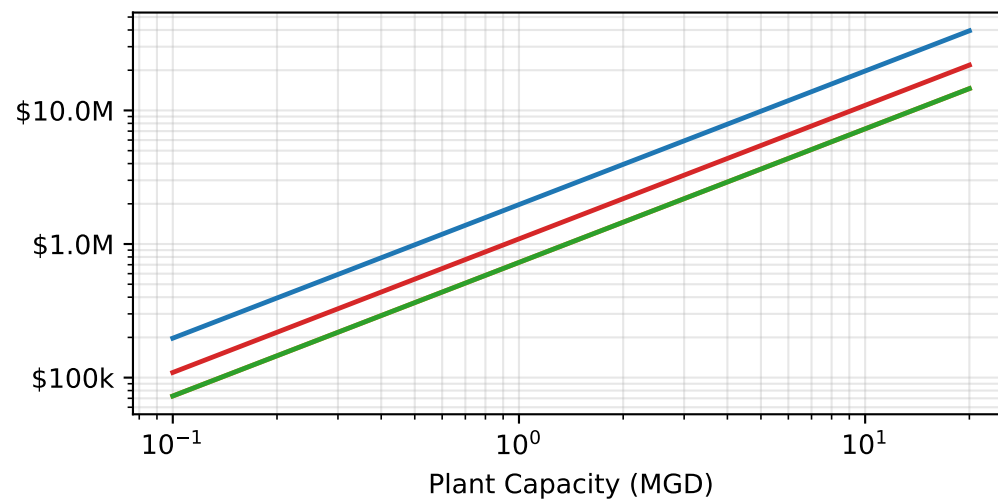
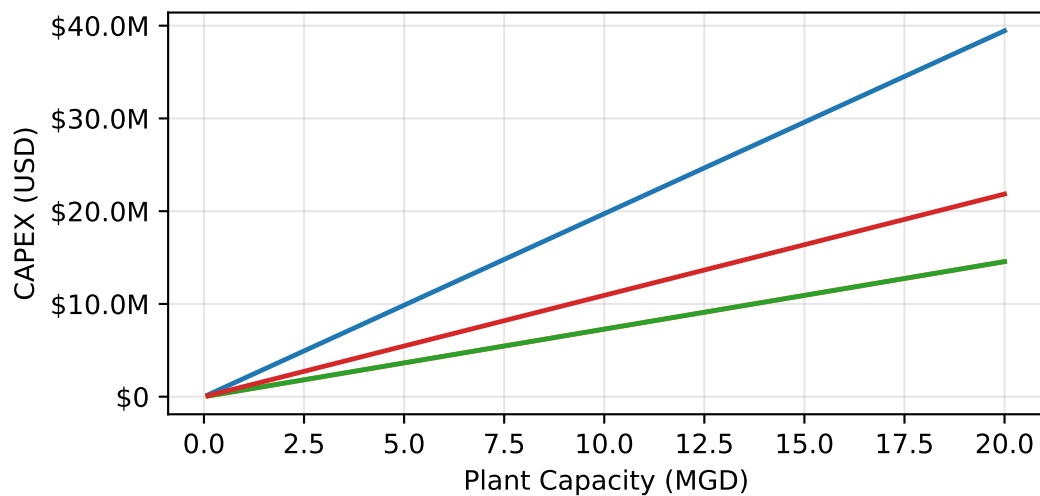
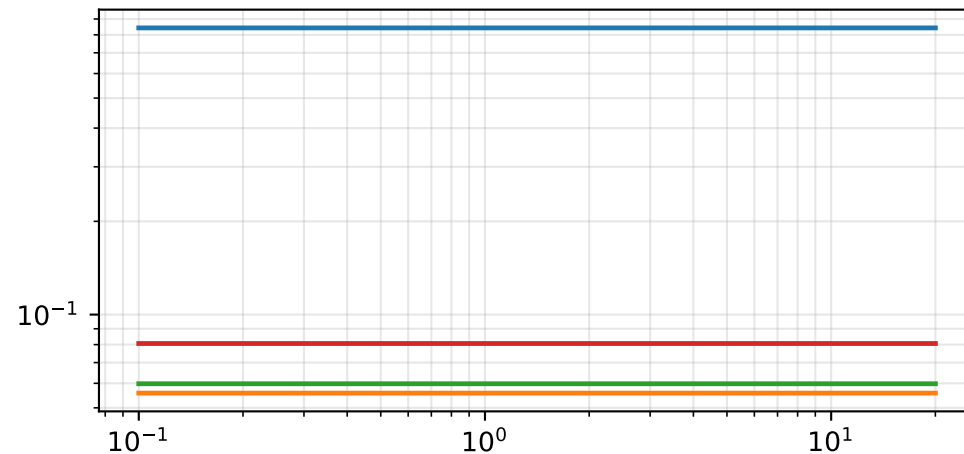
— Anion Exchange — Cation Exchange — GAC - Pressure Vessels — GAC - Gravity Basins

Membrane Processes

Linear Scale



Log-Log Scale

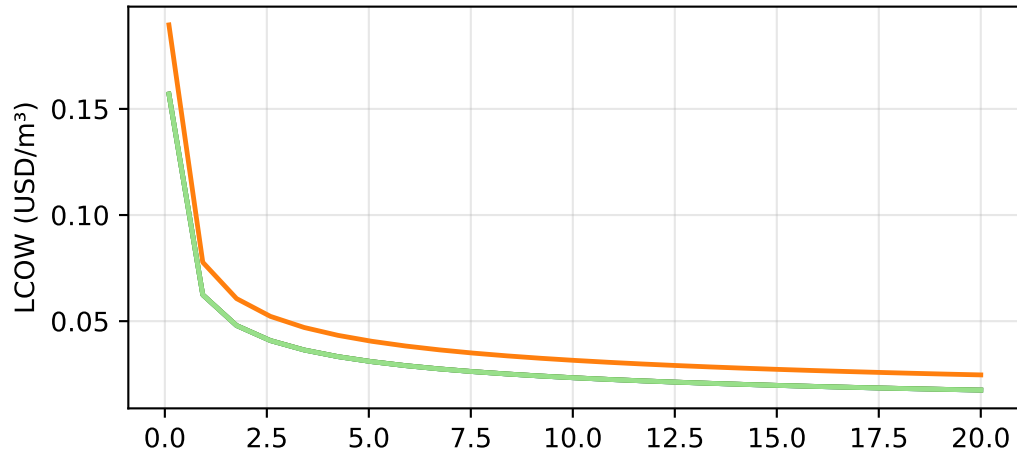


Reverse Osmosis Microfiltration Ultrafiltration Nanofiltration

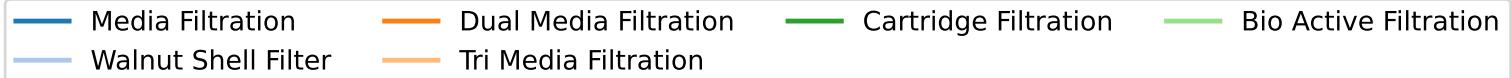
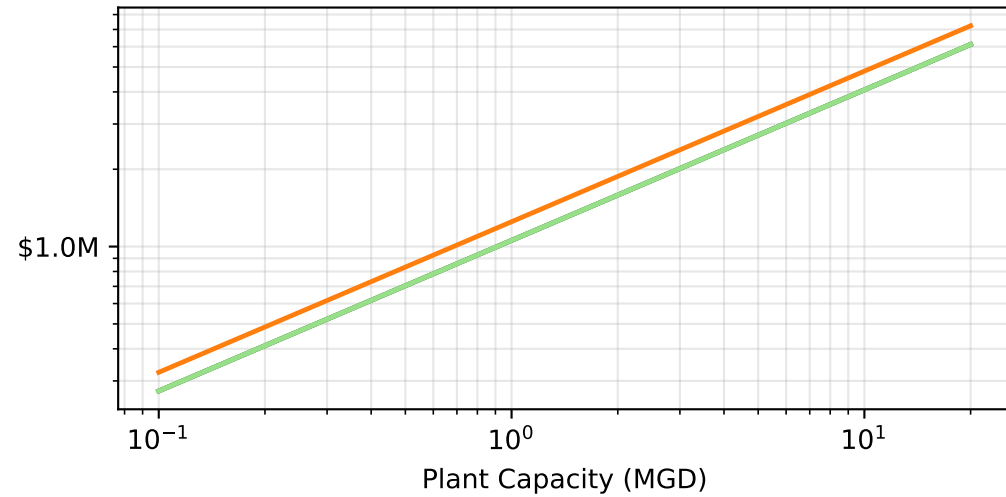
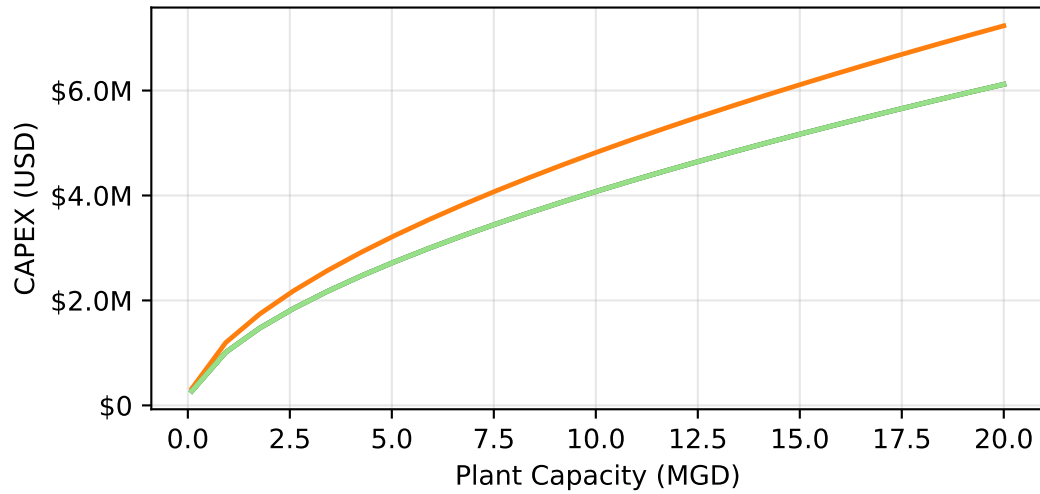
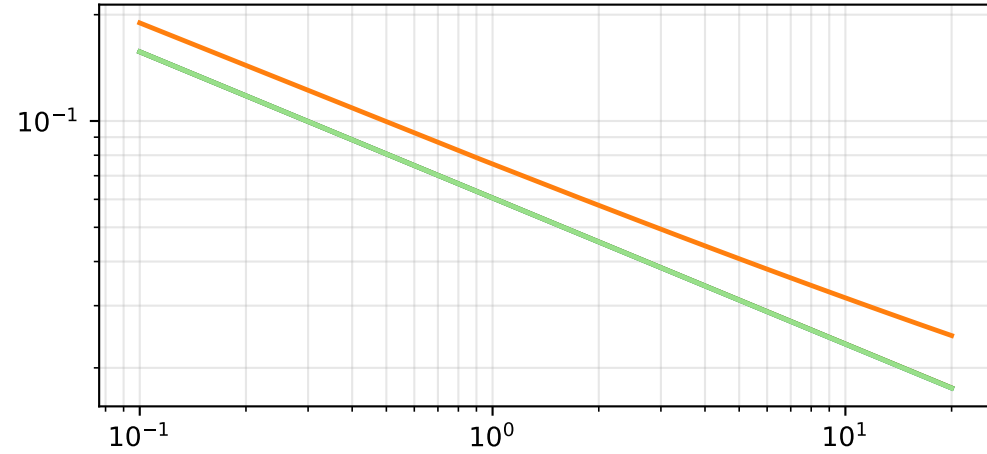


Media Filtration

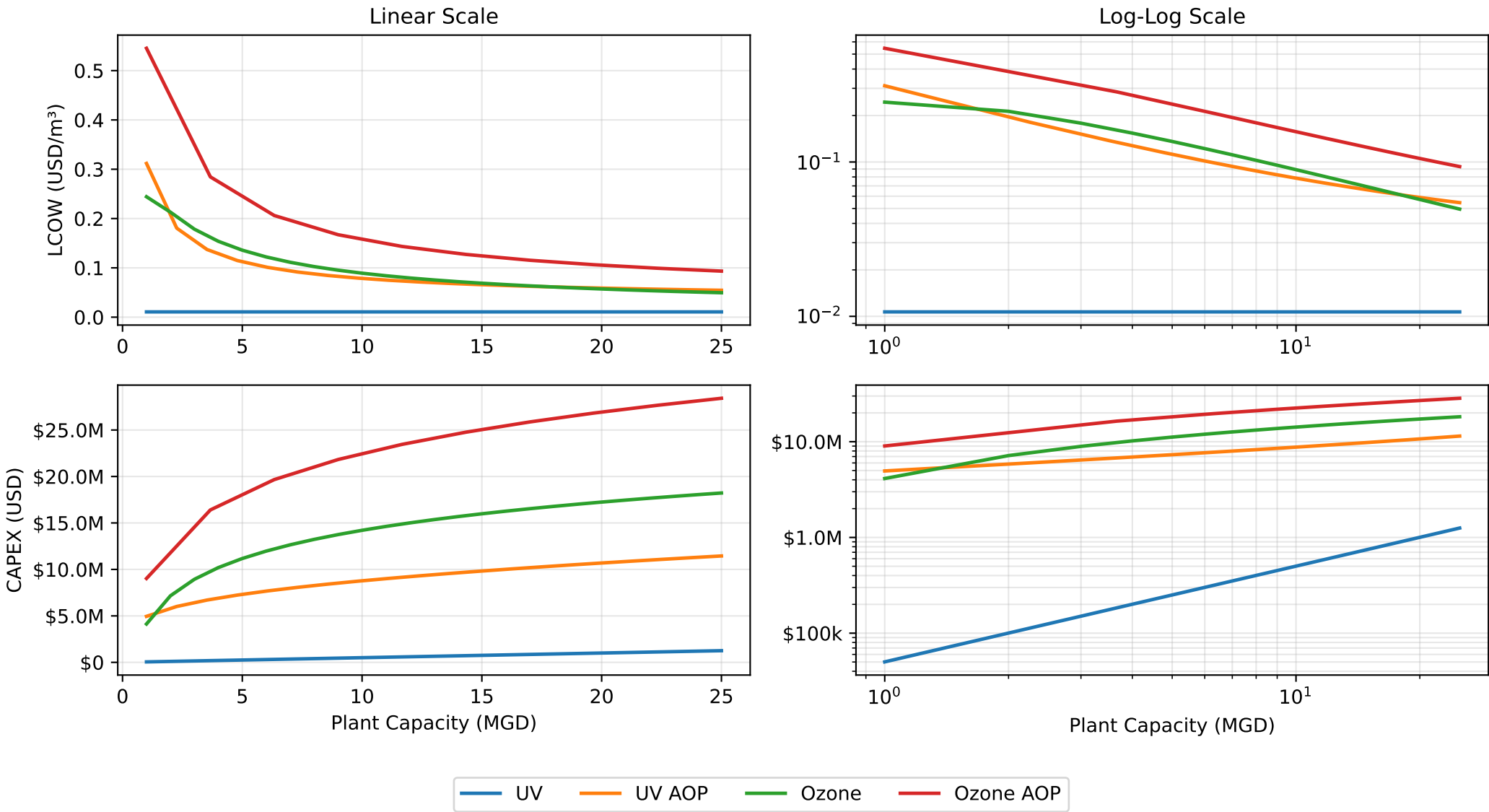
Linear Scale



Log-Log Scale



UV & Ozone AOP



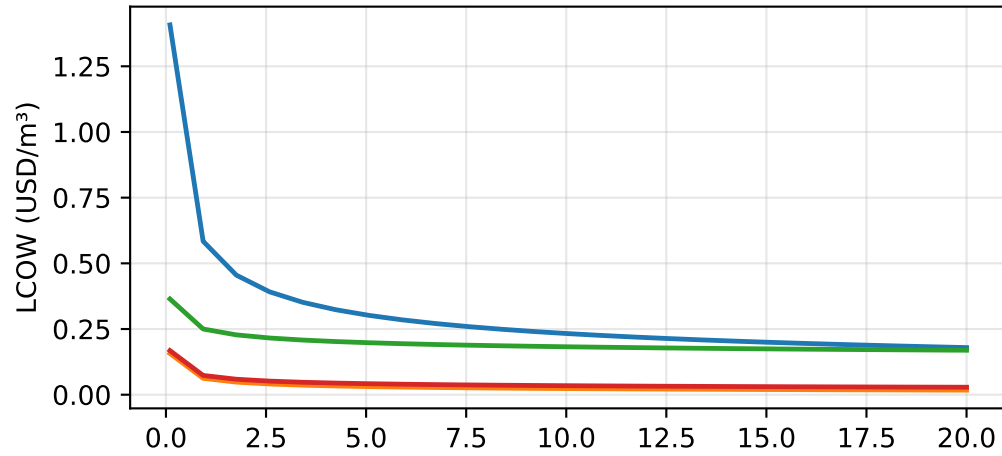
Details:

UV and ozone AOP costs are run with a fixed oxidant dose of 10 mg/L and UV SEC of 0.1 kWh/m³.

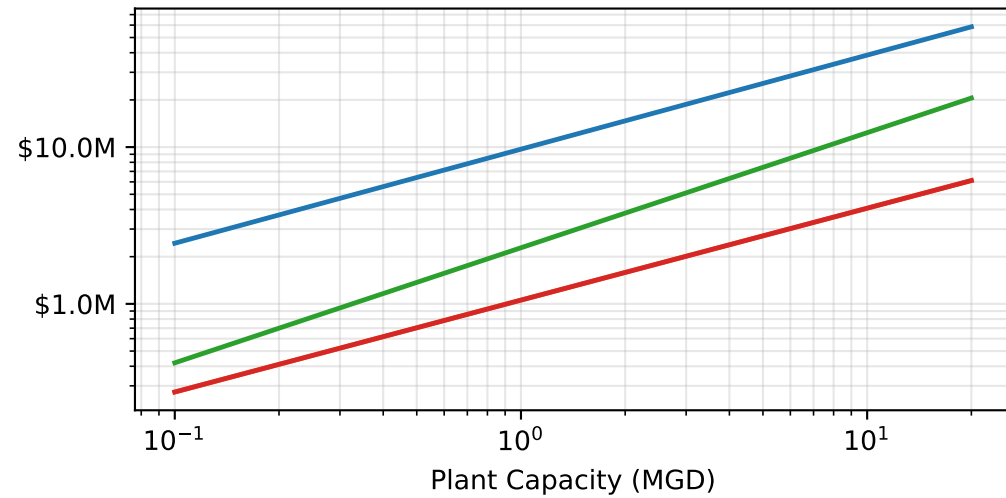
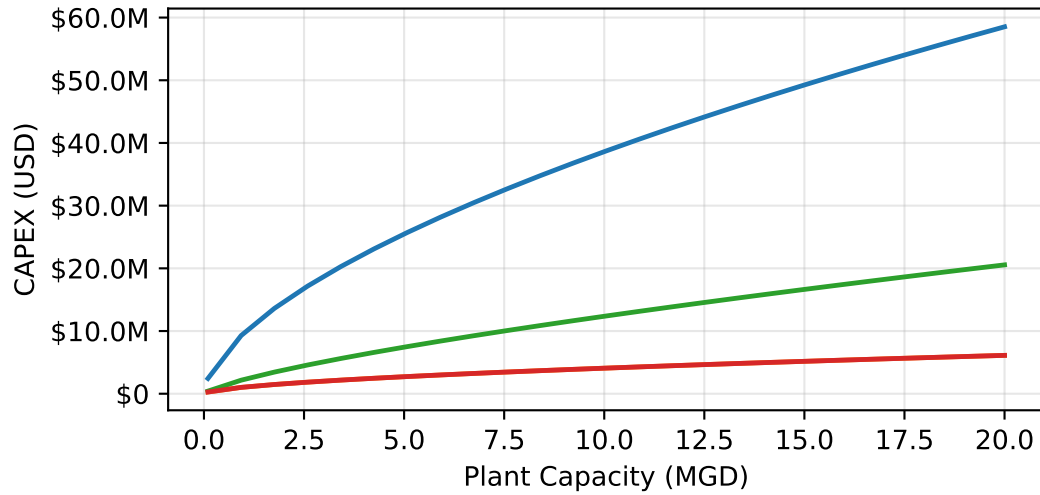
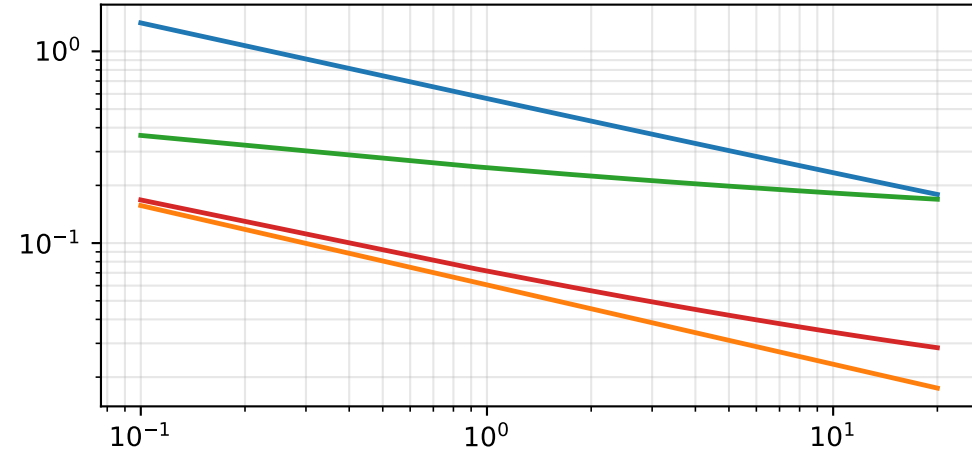
Ozone costs are for a fixed ozone dose of 9 mg/L.

Biological Treatment

Linear Scale



Log-Log Scale

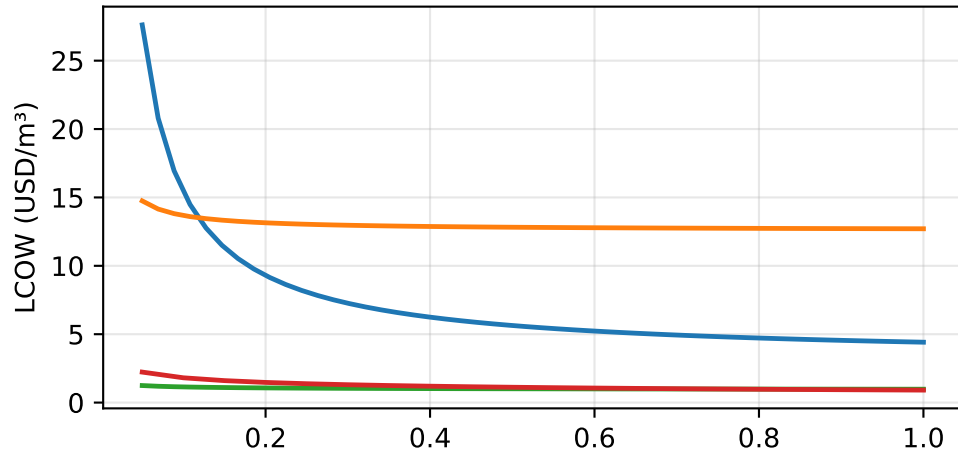


— Anaerobic Digestion Oxidation
 — Bio Active Filtration
 — Fixed Bed
 — Conventional Activated Sludge

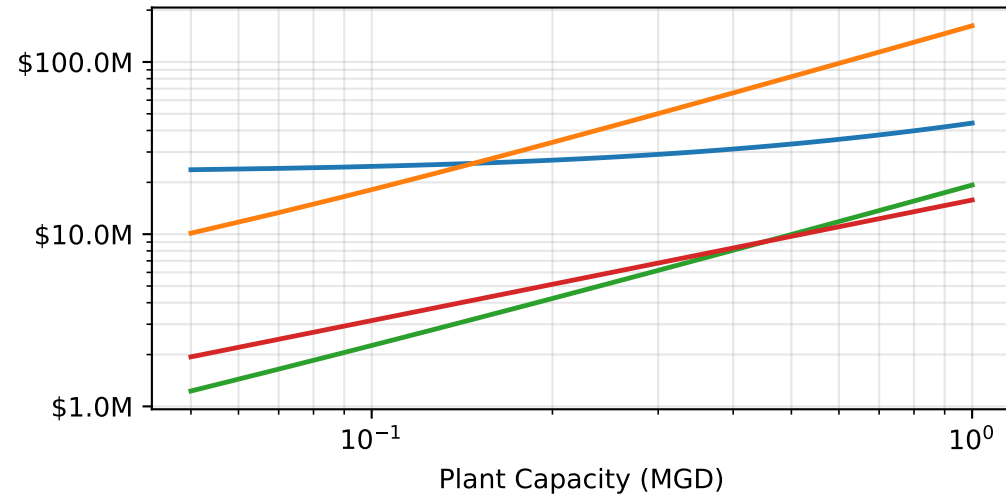
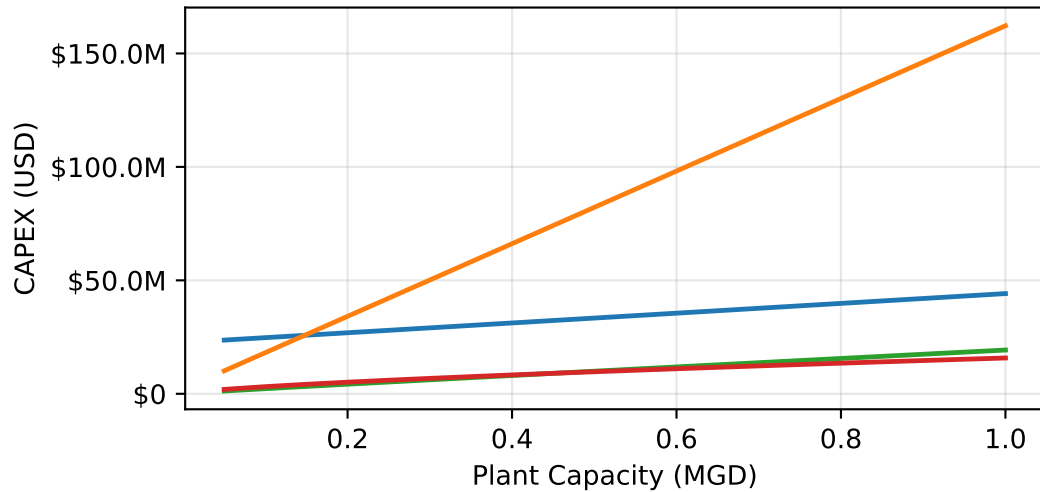
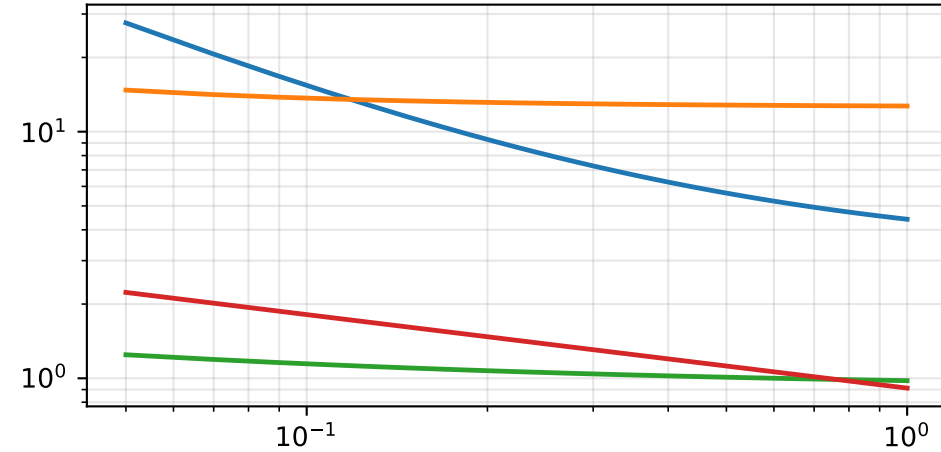


Brine Management

Linear Scale



Log-Log Scale



— Brine Concentrator — Crystallizer — Evaporation Pond — Deep Well Injection

Details:

Brine concentrator and crystallizer CAPEX and SEC are a function inlet flow rate and salinity.

These results are for a feed salinity of 60,000 mg/L TDS for the concentrator and 145,000 mg/L TDS for the crystallizer.

Evaporation pond costs are a function of inlet flow rate and solar radiation, with results shown for 25 MJ/m²/day.

Deep well injection costs are a function of inlet flow rate and pipe distance, with results shown for a pipe distance of 25 miles.

